Nonlinear Hamiltonian analysis of the novel Poincaré gauge theories

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I. INTRODUCTION

When in the course of Human events...

Choice of covariant derivative

$$D_\alpha \phi \equiv (\partial_\alpha + \tfrac{1}{2} A^{ij}{}_\alpha \Sigma_{ij})\phi \tag{1}$$

Field strengths

$$\mathcal{R}^{ij}{}_{kl} \equiv 2h_k{}^\mu h_l{}^\nu \big(\partial_{[\mu} A^{ij}{}_{\nu]} + A^i{}_{o[\mu} A^{oj}{}_{\beta]}\big), \tag{2a}$$

$$\mathcal{T}^i_{kl} \equiv 2h_k{}^\mu h_l{}^\nu \big(\partial_{[\mu} b^i{}_{\nu]} + A^i{}_{o[\mu} b^o{}_{\nu]}\big). \tag{2b}$$

Break quadratic action down using irreducible projection operators (IPOs) with respect to SO(1,3) – six for curvature and three for torsion. We write this as

$$\begin{aligned} L_{\mathrm{T}} = &\sum_{I=1}^{3} m_{\mathrm{p}}{}^2 \hat{\beta}_I \mathcal{T}^i_{jk}{}^I \mathcal{P}_i{}^{jk}{}_l{}^{nm} \mathcal{T}^l_{nm} \\ &+ \sum_{I=1}^{6} \hat{\alpha}_I \mathcal{R}^{ij}{}_{kl}{}^I \mathcal{P}_{ij}{}^{kl}{}_{nm}{}^{pq} \mathcal{R}^{nm}{}_{pq} + L_{\mathrm{m}} \end{aligned} \tag{3}$$

We will use Roman and Greek indices from the middle of the alphabet $i, j\ldots$ $\mu, \nu\ldots$ to refer to general Lorentz and coordinate indices running from 0 to 3, while $a, b\ldots$ and $\alpha, \beta\ldots$ strictly run from 1 to 3. Note this contrasts with our previous notation [1, 2].

II. CONSTRAINED HAMILTONIAN

A. Primary constraints and $3+1$

In order to transition to the Hamiltonian picture, we first define the *canonical* momenta from the Lagrangian

$$\pi_i{}^\mu \equiv \frac{\partial bL}{\partial \partial_0 b^i{}_\mu}, \quad \pi_{ij}{}^\mu \equiv \frac{\partial bL}{\partial \partial_0 A^{ij}{}_\mu}. \tag{4}$$

Since the field strengths (2a) and (2b) make no reference to the velocities of $b^k{}_0$ and $A^{ij}{}_0$, the definitions (4) incur 10 primary constraints

$$\varphi_k{}^0 \equiv \pi_k{}^0 \approx 0, \quad \varphi_{ij}{}^0 \equiv \pi_{ij}{}^0 \approx 0, \tag{5}$$

so that the conjugate fields $b^k{}_0$ and $A^{ij}{}_0$ are non-physical. The constraints (5) are a consequence of Poincaré gauge symmetry; their presence is independent of the couplings, and they are first-class (FC). We refer to them as 'sure' primary first class (sPFC) constraints. In order to systematically isolate the 'sure' non-physical fields, we introduce the $3+1$ (ADM) splitting of the spacetime, in which a spacelike foliation is characterised by unit normal n_k. Any vector which refers to the local Lorentz basis may be split into perpendicular and parallel components $V_i = n_i V_\perp + V_{\bar{i}}$ with respect to this normal. Parallel indices are always denoted with an overbar. In what follows, it is very useful to note the identities $b^{\bar{k}}{}_\alpha h_{\bar{l}}{}^\alpha = \delta^{\bar{k}}_{\bar{l}}$ and $b^{\bar{k}}{}_\alpha h_{\bar{k}}{}^\beta = \delta^\beta_\alpha$. The *lapse* function and *shift* vector are defined with reference to the non-physical part of the translational gauge field

$$N \equiv n_k b^k{}_0, \quad N^\alpha \equiv h_{\bar{k}}{}^\alpha b^k{}_0. \tag{6}$$

The remaining momenta are expressed in the *parallel* form $\hat{\pi}_i{}^{\bar{k}} \equiv \pi_i{}^\alpha b^k{}_\alpha$ and $\hat{\pi}_{ij}{}^{\bar{k}} \equiv \pi_{ij}{}^\alpha b^k{}_\alpha$. From the irreducible expansion of the Lagrangian (3), the parallel momenta can be expressed as functions of the field strengths

$$J^{-1}\hat{\pi}_i{}^{\bar{k}} = \frac{\partial L}{\partial \mathcal{T}^i_{\perp\bar{k}}} = 4m_{\mathrm{p}}{}^2 \sum_{I=1}^{3} \hat{\beta}_I{}^I \mathcal{P}_i{}^{\perp\bar{k}}{}_n{}^{mo} \mathcal{T}^n{}_{mo}\,, \tag{7a}$$

$$J^{-1}\hat{\pi}_{ij}{}^{\bar{k}} = \frac{\partial L}{\partial \mathcal{R}^{ij}{}_{\perp\bar{k}}} = 8\sum_{I=1}^{6} \hat{\alpha}_I{}^I \mathcal{P}_{ij}{}^{\perp\bar{k}}{}_{mn}{}^{pq} \mathcal{R}^{mn}{}_{pq}, \tag{7b}$$

where the measure $J = b/N$ on the foliation is strictly physical. Within the field strengths, the $3+1$ splitting is used again to separate out the 'sure' non-physical fields and non-canonical velocities

$$\mathcal{T}^i_{kl} = \mathcal{T}^i_{\bar{k}\bar{l}} + 2n_{[k}\mathcal{T}^i_{\perp\bar{l}]}\,, \tag{8a}$$

$$\mathcal{R}^{ij}{}_{kl} = \mathcal{R}^{ij}{}_{\bar{k}\bar{l}} + 2n_{[k}\mathcal{R}^{ij}{}_{\perp\bar{l}]}, \tag{8b}$$

where such variables are confined to the second term in each case.

Even within the definitions of the parallel momenta, it is generically possible to identify further primary constraints. Beginning with (7a), we find that the 12 translational parallel momenta decompose into 4 irreducible representations of O(3) as follows

$$\hat{\pi}_{k\bar{l}} = \hat{\pi}_{\bar{k}\bar{l}} + n_k \hat{\pi}_{\perp\bar{l}}, \tag{9a}$$

$$\hat{\pi}_{\bar{k}\bar{l}} = \frac{1}{3}\eta_{\bar{k}\bar{l}}\hat{\pi} + \hat{\hat{\pi}}_{\bar{k}\bar{l}} + \tilde{\hat{\pi}}_{\bar{k}\bar{l}}. \tag{9b}$$

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TABLE I. f

#	criticality equalities	unitary inequalities	0^-	0^+	1^-	1^+	2^-	2^+	D.o.F
Case 3	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_4 = \hat{\alpha}_6 = \hat{\beta}_2 = \hat{\beta}_1 + 2\hat{\beta}_3 = 0$	$\hat{\alpha}_3 < 0 \wedge \hat{\alpha}_5 < 0 \wedge \hat{\beta}_1 < 0$	●		○ ○		○	○ ○	● ○ ○
Case 17	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_3 = \hat{\alpha}_4 = \hat{\alpha}_6 = \hat{\beta}_2 = \hat{\beta}_1 + 2\hat{\beta}_3 = 0$	$\hat{\alpha}_5 < 0$	○		○ ○		○	○ ○	○ ○
Case 20	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_4 = \hat{\alpha}_5 = \hat{\alpha}_6 = 0$	$0 < \hat{\beta}_3 \wedge \hat{\alpha}_3 < 0$	●	○ ○	○ ○	○ ○	○	○ ○	
Case 24	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_4 = \hat{\alpha}_6 = \hat{\beta}_1 = 0$	$0 < \hat{\beta}_3 \wedge \hat{\alpha}_3 < 0$	●	○ ○	○ ○	○ ○			
Case *5 25	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_4 = \hat{\alpha}_5 = \hat{\alpha}_6 = \hat{\beta}_1 = 0$	$0 < \hat{\beta}_3 \wedge \hat{\alpha}_3 < 0$	●	○ ○	○ ○	○ ○			●
Case *6 26	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_4 = \hat{\alpha}_5 = \hat{\alpha}_6 = \hat{\beta}_1 = \hat{\beta}_2 = 0$	$0 < \hat{\beta}_3 \wedge \hat{\alpha}_3 < 0$	●			○ ○			
Case 28	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_4 = \hat{\alpha}_6 = \hat{\beta}_1 = \hat{\beta}_2 = 0$	$0 < \hat{\beta}_3 \wedge \hat{\alpha}_3 < 0$	●		○	○ ○			
Case 32	$\hat{\alpha}_1 = \hat{\alpha}_2 = \hat{\alpha}_4 = \hat{\alpha}_5 = \hat{\alpha}_6 = \hat{\beta}_2 = 0$	$0 < \hat{\beta}_3 \wedge \hat{\alpha}_3 < 0$	●		○ ○	○ ○	○	○ ○	

These are 0^+, 1^+, 1^- and 2^- sectors. Applying this decomposition to (7a) as a whole, we obtain 4 functions which, with the aid of (8a), are simultaneously defined both in terms of canonical and non-canonical variables

$$\varphi \equiv J^{-1}\hat{\pi} = \hat{\beta}_2 \eta^{\overline{kl}} \mathcal{T}_{\perp\overline{kl}}\,, \tag{10a}$$

$$\begin{aligned}\hat{\varphi}_{\overline{kl}} &\equiv J^{-1}\hat{\hat{\pi}}_{\overline{kl}} + \frac{1}{3}(\hat{\beta}_1 - \hat{\beta}_3)\mathcal{T}_{\overline{kl}\perp} \\ &= \frac{1}{3}(\hat{\beta}_1 + 2\hat{\beta}_3)\mathcal{T}_{\perp[\overline{kl}]}\,,\end{aligned} \tag{10b}$$

$$\begin{aligned}\varphi_{\perp\overline{k}} &\equiv J^{-1}\hat{\pi}_{\perp\overline{k}} + \frac{1}{3}(\hat{\beta}_1 - \hat{\beta}_2)\vec{\mathcal{T}}_{\overline{k}} \\ &= \frac{1}{3}(2\hat{\beta}_1 + \hat{\beta}_2)\mathcal{T}_{\perp\overline{k}\perp}\,,\end{aligned} \tag{10c}$$

$$\tilde{\varphi}_{\overline{kl}} \equiv J^{-1}\tilde{\hat{\pi}}_{\overline{kl}} = \hat{\beta}_1 \mathcal{T}_{\perp\langle\overline{kl}\rangle}\,, \tag{10d}$$

where $\vec{\mathcal{T}}_{\overline{k}} \equiv \mathcal{T}^{\bar{i}}_{\overline{ki}}$ and $\mathcal{T}_{\perp\langle\overline{kl}\rangle} \equiv \mathcal{T}_{\perp(\overline{kl})} - \frac{1}{3}\eta_{\overline{kl}}\eta^{\overline{ij}}\mathcal{T}_{\perp\overline{ij}}$. In each case, if the combination of coupling constants appearing in the non-canonical definition vanishes, the canonically defined function becomes a primary 'if' constraint (PiC). An analogous construction is available for (7b), with the 18 remaining momenta decomposing as follows

$$\hat{\pi}_{kl\bar{o}} = \hat{\pi}_{\overline{klo}} + 2n_{[k}\hat{\pi}_{\perp\bar{l}]\bar{o}}, \tag{11a}$$

$$\hat{\pi}_{\perp\overline{kl}} = \frac{1}{3}\eta_{\overline{kl}}\hat{\pi}_{\perp} + \hat{\hat{\pi}}_{\perp\overline{kl}} + \tilde{\hat{\pi}}_{\perp\overline{kl}}, \tag{11b}$$

$$\hat{\pi}_{\overline{klo}} = \frac{1}{6}\epsilon_{\overline{klo}\perp}\,{}^{\mathrm{P}}\hat{\pi} + \vec{\hat{\pi}}_{[\overline{k}}\eta_{\bar{l}]\bar{o}} + \frac{4}{3}\,{}^{\mathrm{T}}\hat{\pi}_{\overline{klo}}. \tag{11c}$$

These are the 0^+, 1^+ and 2^+, and then 0^-, 1^- and 2^- representations. The 6 PiC functions from (7b) are then

$$\begin{aligned}\varphi_{\perp} &\equiv J^{-1}\hat{\pi}_{\perp} + m_{\mathrm{p}}{}^2(\hat{\alpha}_4 - \hat{\alpha}_6)\,\underline{\mathcal{R}} \\ &= m_{\mathrm{p}}{}^2(\hat{\alpha}_4 + \hat{\alpha}_6)\,\mathcal{R}_{\perp\perp},\end{aligned} \tag{12a}$$

$$\begin{aligned}{}^{\mathrm{P}}\varphi &\equiv J^{-1}\,{}^{\mathrm{P}}\hat{\pi} + m_{\mathrm{p}}{}^2(\hat{\alpha}_2 - \hat{\alpha}_3)\,{}^{\mathrm{P}}\mathcal{R}_{\circ\perp} \\ &= m_{\mathrm{p}}{}^2(\hat{\alpha}_2 + \hat{\alpha}_3)\,{}^{\mathrm{P}}\mathcal{R}_{\perp\circ},\end{aligned} \tag{12b}$$

$$\begin{aligned}\hat{\varphi}_{\perp\overline{kl}} &\equiv J^{-1}\hat{\hat{\pi}}_{\perp\overline{kl}} + m_{\mathrm{p}}{}^2(\hat{\alpha}_2 - \hat{\alpha}_5)\,\underline{\mathcal{R}}_{[\overline{kl}]} \\ &= m_{\mathrm{p}}{}^2(\hat{\alpha}_2 + \hat{\alpha}_5)\,\mathcal{R}_{\perp[\overline{kl}]\perp},\end{aligned} \tag{12c}$$

$$\begin{aligned}\vec{\varphi}_{\overline{k}} &\equiv J^{-1}\vec{\hat{\pi}}_{\overline{k}} + m_{\mathrm{p}}{}^2(\hat{\alpha}_4 - \hat{\alpha}_5)\,\mathcal{R}_{\overline{k}\perp} \\ &= m_{\mathrm{p}}{}^2(\hat{\alpha}_4 + \hat{\alpha}_5)\,\mathcal{R}_{\perp\overline{k}},\end{aligned} \tag{12d}$$

$$\begin{aligned}\tilde{\varphi}_{\perp\overline{kl}} &\equiv J^{-1}\tilde{\hat{\pi}}_{\perp\overline{kl}} + m_{\mathrm{p}}{}^2(\hat{\alpha}_1 - \hat{\alpha}_4)\,\underline{\mathcal{R}}_{\langle\overline{kl}\rangle} \\ &= m_{\mathrm{p}}{}^2(\hat{\alpha}_1 + \hat{\alpha}_4)\,\mathcal{R}_{\perp\langle\overline{kl}\rangle\perp},\end{aligned} \tag{12e}$$

$$\begin{aligned}{}^{\mathrm{T}}\varphi_{\overline{klo}} &\equiv J^{-1}\,{}^{\mathrm{T}}\hat{\pi}_{\overline{klo}} + m_{\mathrm{p}}{}^2(\hat{\alpha}_1 - \hat{\alpha}_2)\,{}^{\mathrm{T}}\mathcal{R}_{\overline{klo}\perp} \\ &= m_{\mathrm{p}}{}^2(\hat{\alpha}_1 + \hat{\alpha}_2)\,{}^{\mathrm{T}}\mathcal{R}_{\perp\overline{klo}},\end{aligned} \tag{12f}$$

where ${}^{\mathrm{P}}\mathcal{R}_{\circ\perp} \equiv \epsilon^{\overline{ijk}\perp}\mathcal{R}_{\overline{ijk}\perp}$, ${}^{\mathrm{P}}\mathcal{R}_{\perp\circ} \equiv \epsilon^{\overline{ijk}\perp}\mathcal{R}_{\perp\overline{ijk}}$, $\underline{\mathcal{R}}_{\overline{kl}} \equiv \mathcal{R}^{\bar{i}}{}_{\overline{kli}}$ and $\underline{\mathcal{R}} \equiv \underline{\mathcal{R}}^{\bar{i}}{}_{\bar{i}}$. By this analysis, the possible occurance of primary constraints is systematically exhausted.

B. Secondary constraints and the Hamiltonian

In order to be consistent, a primary constraint should not have any velocity within the final mass shell, so its commutator with the *total* Hamiltonian should weakly vanish

$$\dot{\varphi}(x_1) \equiv \int \mathrm{d}^3x_2\{\varphi(x_1), \mathcal{H}_{\mathrm{T}}(x_2)\} \approx 0. \tag{13}$$

The the total Hamiltonian is related to the *canonical* Hamiltonian, the Legendre-transformed Lagrangian, by the constraints and their multiplier fields

$$\mathcal{H}_{\mathrm{T}} \equiv \mathcal{H}_{\mathrm{C}} + u^k{}_0\varphi_k{}^0 + \frac{1}{2}u^{ij}{}_0\varphi_{ij}{}^0 + (u\cdot\varphi), \tag{14}$$

where the last term schematically represents any PICs which may arise. The canonical Hamiltonian may generally be collected into the insightful *Dirac* form

$$\mathcal{H}_{\rm C} \equiv N\mathcal{H}_\perp + N^\alpha \mathcal{H}_\alpha - \frac{1}{2}A^{ij}{}_0\mathcal{H}_{ij} + \partial_\alpha \mathscr{D}^\alpha, \tag{15}$$

i.e. as a linear function of the non-physical fields up to a surface term. The remaining functions which appear in (15) are

$$\mathcal{H}_\perp \equiv \hat{\pi}_i{}^{\bar{k}}\mathcal{T}^i_{\perp\bar{k}} + \frac{1}{2}\hat{\pi}_{ij}{}^{\bar{k}}\mathcal{R}^{ij}{}_{\perp\bar{k}} - JL - n^k D_\alpha \pi_k{}^\alpha, \tag{16a}$$

$$\mathcal{H}_\alpha \equiv \pi_i{}^\beta T^i_{\alpha\beta} + \frac{1}{2}\pi_{ij}{}^\beta R^{ij}{}_{\alpha\beta} - b^k{}_\alpha D_\beta \pi_k{}^\beta, \tag{16b}$$

$$\mathcal{H}_{ij} \equiv 2\pi_{[i}{}^\alpha b_{j]\alpha} - D_\alpha \pi_{ij}{}^\alpha, \tag{16c}$$

$$\mathscr{D}^\alpha \equiv b^i{}_0 \pi_i{}^\alpha + \frac{1}{2}A^{ij}{}_0\pi_{ij}{}^\alpha. \tag{16d}$$

It is clear from (20) and the Dirac form (15) that the consistency of (5) invokes 10 'sure' *secondary* first class (sSFC) constriants

$$\mathcal{H}_\perp \approx 0, \quad \mathcal{H}_\alpha \approx 0, \quad \mathcal{H}_{ij} \approx 0. \tag{17}$$

It is important to note that while the sSFCs in (17) are always enforced, it does not always follow that 2×10 D.o.F are removed from the theory, as is the case with the sPFCs in (5). The functions involved are quite complicated, and may degenerately express a reduced number of FCs, or FCs which only appear at deeper levels in the consistency chain. Indeed, while this is very rare in the literature, we will find that it occurs for all the novel theories, as a consequence of vanishing mass parameters.

The linear and rotational super-momenta $\mathcal{H}_\alpha$ and $\mathcal{H}_{ij}$ are kinematic generators which do not impinge on the dynamics. In the evaluation of (20), it is sufficient to work purely with the super-Hamiltonian $\mathcal{H}_\perp$, which is readily expanded out using Eqs. (3), (8a) and (8b) to give

$$\begin{aligned}\mathcal{H}_\perp = &\sum_{I=1}^{3} m_{\rm p}{}^2 J\hat{\beta}_I \Big[4\mathcal{T}^i_{\perp\bar{k}}\, {}^I\mathcal{P}_i{}^{\perp\bar{k}}{}_j{}^{\perp\bar{l}}\mathcal{T}^j_{\perp\bar{l}} \\ &- \mathcal{T}^i_{\overline{mk}}\, {}^I\mathcal{P}_i{}^{\overline{mk}}{}_j{}^{\overline{nl}}\mathcal{T}^j_{\overline{nl}}\Big] \\ &+ \sum_{I=1}^{6} J\hat{\alpha}_I \Big[4\mathcal{R}^{ip}{}_{\perp\bar{k}}\, {}^I\mathcal{P}_{ip}{}^{\perp\bar{k}}{}_{jq}{}^{\perp\bar{l}}\mathcal{R}^{jq}{}_{\perp\bar{l}} \\ &- \mathcal{R}^{ip}{}_{\overline{mk}}\, {}^I\mathcal{P}_{ip}{}^{\overline{mk}}{}_{jq}{}^{\overline{nl}}\mathcal{R}^{jq}{}_{\overline{nl}}\Big] - n^k D_\alpha \pi_k{}^\alpha.\end{aligned} \tag{18}$$

The non-canonical 'perpendicaular' field strengths appearing in (18) may be canonicalised at length by the dual PIC definitions in Eqs. (10a) to (10d) and Eqs. (12a) to (12f), resulting in terms quadratic in the PIC functions, and in the canonical 'parallel' field strengths. The resulting expression is quite lengthy, but can be simplified for any given theory by safely elimiating those PIC functions which become constraints. When evaluating

TABLE II. Spin-parity sectors and associated PiCs, along with their kinetic and mass parameters.

J^P	PiC	Kinetic parameter	Mass parameter
0^+	φ $\varphi_\perp$	$\hat{\beta}_2$ $\hat{\alpha}_4 + \hat{\alpha}_6$	$\hat{\alpha}_0(2\hat{\alpha}_0 + \hat{\beta}_2)$
0^-	${}^{\rm P}\varphi$	$\hat{\alpha}_2 + \hat{\alpha}_3$	$\hat{\alpha}_0 + 2\hat{\beta}_3$
1^+	$\hat{\varphi}_{\overline{kl}}$ $\hat{\varphi}_{\perp\overline{kl}}$	$\hat{\beta}_1 + 2\hat{\beta}_3$ $\hat{\alpha}_2 + \hat{\alpha}_5$	$(\hat{\alpha}_0 + 2\hat{\beta}_3)(\hat{\alpha}_0 - \hat{\beta}_1)$
1^-	$\varphi_{\perp\bar{k}}$ $\varphi_{\bar{k}}$	$2\hat{\beta}_1 + \hat{\beta}_2$ $\hat{\alpha}_4 + \hat{\alpha}_5$	$(2\hat{\alpha}_0 + \hat{\beta}_2)(\hat{\alpha}_0 - \hat{\beta}_1)$
2^+	$\tilde{\varphi}_{\overline{kl}}$ $\tilde{\varphi}_{\perp\overline{kl}}$	$\hat{\beta}_1$ $\hat{\alpha}_1 + \hat{\alpha}_4$	$\hat{\alpha}_0(\hat{\alpha}_0 - \hat{\beta}_1)$
2^-	${}^{\rm T}\varphi_{\overline{klo}}$	$\hat{\alpha}_1 + \hat{\alpha}_2$	$\hat{\alpha}_0 - \hat{\beta}_1$

TABLE III. Nonstandard abbreviations.

D.o.F	degrees of freedom
PPM	primary Poisson matrix
P/S/TiC	primary/secondary/tertiary if-constraint
F/SC	first/second-class
iP/S/TF/SC	P/S/TiC which is F/SC on the final shell
sP/SFC	sure primary/secondary constraint, always FC

Poisson brackets in general, we note the following very useful identities

$$\begin{aligned}\frac{\partial n_l}{\partial b^k{}_\mu} &= -n_k h_{\bar{l}}{}^\mu, \quad \frac{\partial h_l{}^\nu}{\partial b^k{}_\mu} = -h_k{}^\nu h_l{}^\mu, \\ \frac{\partial b}{\partial b^k{}_\nu} &= b h_k{}^\nu, \quad \frac{\partial J}{\partial b^k{}_\nu} = J h_{\bar{k}}{}^\nu.\end{aligned} \tag{19}$$

The consistency of the PICs is less straightforward. Generally, the PICs may be FC or SC within their own mass shell. In the case that a PIC is FC, (20) provides a secondary if-constraint (SIC). Possibly, the PIC and SIC do not commute; in that case both become SC within the new mass shell and the consistency of the SIC allows a multiplier to be determined

$$\begin{aligned}\dot{\chi}(x_1) \equiv \int {\rm d}^3x_2 N\{\chi(x_1), \mathcal{H}_\perp(x_2)\} \\ + u\cdot\{\chi(x_1), \varphi(x_2)\} \approx 0.\end{aligned} \tag{20}$$

Otherwise, a tertiary if-constraint (TIC) may be found, and the process continues until the contraint chain from the PIC is absorbed by another chain, or by the sSFCs. In the case that a PIC is already SC within the PIC mass shell, its chain terminates immediately and two multipliers are determined. We note that occasionally, a constraint may be encountered at some deep level which retroactively terminates the chain at a shallower point. Only once the algoritm has terminated is it safe to categorise the if-constraints as FC or SC.

In the case of the linearised theory,

III. MASSIVE-ONLY THEORIES

The simplest theories (constraints are only functions of momenta) turn out to all be massive-only in the weak regime. Obviously this is quite useless, but we can use it as a staging area for the supposedly viable theories. We construct the Poisson matrices of primary if-constraints (PICs) to be strictly on-shell. The entries are purely schematic, and represent some usually long expressions which currently exist only on paper. As expected, all nonzero Poisson brackets for the selected cases are linear in the momenta. The labels around the edge give the O(3) irreps of the constraint fields, with multiplicities. For ease, the matrices are divided into quadrants (should more than one quadrant exist) for translational, rotational and mixed brackets. The mixed brackets are useful for chain termination.

A. Case $^{*6}26$

The nonlinear PPM of Case $^{*6}26$ takes the form:

$$\begin{array}{c|cccccccc|c}
 & \varphi & \varphi_{\perp\overline{k}} & \widetilde{\varphi}_{\overline{kl}} & \varphi_{\perp} & \hat{\varphi}_{\perp\overline{kl}} & \vec{\varphi}_{\overline{k}} & \widetilde{\varphi}_{\perp\overline{kl}} & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \\
\hline
\varphi & \cdot & \cdot & \cdot & \boxed{\cdot} & \cdot & \cdot & \cdot & \cdot & 1 \\
\varphi_{\perp\overline{k}} & \cdot & \hat{\pi} & \cdot & \cdot & \hat{\pi} & \boxed{\cdot} & \cdot & \cdot & 3 \\
\widetilde{\varphi}_{\overline{kl}} & \cdot & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \boxed{\cdot} & \hat{\pi} & 5 \\
\varphi_{\perp} & \boxed{\cdot} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 \\
\hat{\varphi}_{\perp\overline{kl}} & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 3 \\
\vec{\varphi}_{\overline{k}} & \cdot & \boxed{\cdot} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 3 \\
\widetilde{\varphi}_{\perp\overline{kl}} & \cdot & \cdot & \boxed{\cdot} & \cdot & \cdot & \cdot & \cdot & \cdot & 5 \\
{}^{\mathrm{T}}\varphi_{\overline{klo}} & \cdot & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & \cdot & 5 \\
\hline
 & 1 & 3 & 5 & 1 & 3 & 3 & 5 & 5 &
\end{array} \tag{21}$$

The elements of the matrix schematically represent the nonlinear Poisson brackets between the PiCs. The PiCs are labelled, along with their multiplicities, at the edges of the PPM. They are arranged so as to divide the matrix into translational and rotational blocks, separated by (+). All brackets are restricted to the PiC shell. Commuting PiCs are dentoted by $(\cdot)$. Non-commuting PiCs $(\hat{\pi})$ are strictly linear combinations of $\hat{\pi}_{ij}{}^{\overline{k}}$ and $\hat{\pi}_{i}{}^{\overline{k}}$. Generally, these expressions can be quite lengthy, so we confine them to Appendix B. Constant terms only arise in brackets between conjugate PiCs ($\boxed{\cdot}$), and then only if both PiCs have non-vanishing mass parameters. Since all the PiC mass parameters vanish in Case $^{*6}26$, no constant terms can arise. The linearised theory is sensitive only to these constant terms, but we see from (21) that the conjugate PiCs also commute in the nonlinear Case $^{*6}26$.

Let us now consider the consistency of the PiCs, and implement the algorithm. Within the PiC shell, we encounter the following SiCs

$$\chi^{\flat}{}_{\perp\overline{k}} \approx -2\eta^{\flat\overline{ml}}\mathcal{D}^{\flat}{}_{\overline{m}}\hat{\hat{\pi}}^{\flat}{}_{\overline{kl}}, \tag{22a}$$

$$\hat{\chi}^{\flat}{}_{\perp\overline{kl}} \approx \hat{\hat{\pi}}^{\flat}{}_{\overline{kl}} - \frac{1}{12}\epsilon^{\flat}{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^{\flat}{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^{\flat}. \tag{22b}$$

Also within this shell, we find $\mathcal{H}^{\flat}{}_{\perp}$ and $\mathcal{H}^{\flat}{}_{\perp\overline{k}}$ already vanish weakly, while the linear super-momentum and vector part of the rotational super-momentum give further sSFCs

$$\mathcal{H}^{\flat}{}_{\alpha} \approx -h^{\flat}{}_{\alpha}{}^{\overline{j}}\eta^{\flat\overline{kl}}\mathcal{D}^{\flat}{}_{\overline{k}}\hat{\hat{\pi}}^{\flat}{}_{\overline{jl}}, \tag{23a}$$

$$\mathcal{H}^{\flat}{}_{\overline{kl}} \approx 2\hat{\hat{\pi}}^{\flat}{}_{\overline{kl}} - \frac{1}{6}\epsilon^{\flat}{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^{\flat}{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^{\flat}. \tag{23b}$$

The SICs clearly vanish in the sSFC sub-shell, terminating the algorithm immediately. We find that $\mathcal{H}^{\flat}{}_{\alpha}$ is already implied by $\mathcal{H}^{\flat}{}_{\overline{kl}}$, which constitutes a total of 3 sSFCs. The PiCs are all FC, and the total number of iPFCs can be read off from (21). Recalling also the 10 sPFCs, and counting all FCs twice, we find that there is only 1 propagating D.o.F, as expected from Table I

$$\begin{aligned} 1 = \frac{1}{2}\big(&80 - 2\times 10[\text{sPFC}] - 2\times 3[\text{sSFC}] \\ &- 2\times(1+3+5+1+3+3+5+5)[\text{iPFC}]\big). \end{aligned} \tag{24}$$

B. Case 28

The nonlinear PPM of Case 28 takes the form:

$$\begin{array}{c|cccccc|c}
 & \varphi & \varphi_{\perp\overline{k}} & \widetilde{\varphi}_{\overline{kl}} & \varphi_{\perp} & \widetilde{\varphi}_{\perp\overline{kl}} & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \\
\hline
\varphi & \cdot & \cdot & \cdot & \boxed{\cdot} & \cdot & \cdot & 1 \\
\varphi_{\perp\overline{k}} & \cdot & \hat{\pi} & \cdot & \hat{\pi} & \hat{\pi} & \hat{\pi} & 3 \\
\widetilde{\varphi}_{\overline{kl}} & \cdot & \cdot & \hat{\pi} & \cdot & \boxed{\hat{\pi}} & \hat{\pi} & 5 \\
\varphi_{\perp} & \boxed{\cdot} & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 1 \\
\widetilde{\varphi}_{\perp\overline{kl}} & \cdot & \hat{\pi} & \boxed{\hat{\pi}} & \cdot & \cdot & \cdot & 5 \\
{}^{\mathrm{T}}\varphi_{\overline{klo}} & \cdot & \hat{\pi} & \hat{\pi} & \cdot & \cdot & \cdot & 5 \\
\hline
 & 1 & 3 & 5 & 1 & 5 & 5 &
\end{array} \tag{25}$$

Within the PiC shell, we find that $\varphi^{\flat}$ and $\widetilde{\varphi}^{\flat}{}_{\overline{kl}}$ already weakly vanish, leaving the following SiCs

$$\chi^{\flat}{}_{\perp\overline{k}} \approx -2\eta^{\flat\overline{ml}}\mathcal{D}^{\flat}{}_{\overline{m}}\hat{\hat{\pi}}^{\flat}{}_{\overline{kl}}, \tag{26a}$$

$$\chi^{\flat}{}_{\perp} \approx -\frac{1}{2}\eta^{\flat\overline{ml}}\mathcal{D}^{\flat}{}_{\overline{m}}\vec{\hat{\pi}}^{\flat}{}_{\overline{l}}, \tag{26b}$$

$$\widetilde{\chi}^{\flat}{}_{\perp\overline{kl}} \approx \frac{1}{4}\mathcal{D}^{\flat}{}_{\langle\overline{k}}\vec{\hat{\pi}}^{\flat}{}_{\overline{l}\rangle}, \tag{26c}$$

$$\begin{aligned} {}^{\mathrm{T}}\chi^{\flat}{}_{\overline{klo}} \approx{}& \frac{1}{4}\mathcal{D}^{\flat}{}_{\overline{o}}\hat{\hat{\pi}}^{\flat}{}_{\perp\overline{kl}} + \frac{1}{4}\mathcal{D}^{\flat}{}_{[\overline{l}}\hat{\hat{\pi}}^{\flat}{}_{\perp\overline{k}]\overline{o}} \\ &+ \frac{3}{8}\eta^{\flat}{}_{\overline{o}[\overline{k}|}\eta^{\flat\overline{ij}}\mathcal{D}^{\flat}{}_{\overline{i}}\hat{\hat{\pi}}^{\flat}{}_{\perp|\overline{l}]\overline{j}}, \end{aligned} \tag{26d}$$

which do not give rise to any TiCs. Also within the PiC shell, the following sSFCs appear

$$\mathcal{H}^{\flat}{}_{\alpha} \approx -h^{\flat}{}_{\alpha}{}^{\overline{j}}\eta^{\flat\overline{kl}}\mathcal{D}^{\flat}{}_{\overline{k}}\hat{\hat{\pi}}^{\flat}{}_{\overline{jl}}, \tag{27a}$$

$$\mathcal{H}^{\flat}{}_{\overline{kl}} \approx 2\hat{\hat{\pi}}^{\flat}{}_{\overline{kl}} - \frac{1}{6}\epsilon^{\flat}{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^{\flat}{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^{\flat} + \mathcal{D}^{\flat}{}_{[\overline{k}}\vec{\hat{\pi}}^{\flat}{}_{\overline{l}]}, \tag{27b}$$

$$\mathcal{H}^{\flat}{}_{\perp\overline{k}} \approx \eta^{\flat\overline{jl}}\mathcal{D}^{\flat}{}_{\overline{j}}\hat{\hat{\pi}}^{\flat}{}_{\perp\overline{kl}}. \tag{27c}$$

In this case it is easiest to restrict to sub-shells using the SiCs and sSFCs simultaneously. We first note that

$\mathcal{H}^\flat{}_{\perp\overline{k}}$ restricts $\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}}$ to be solenoidal, dual to the gradient of a scalar, and thus eliminates 2 D.o.F. The remaining D.o.F is eliminated by ${}^{\mathrm{T}}\chi^\flat{}_{\overline{klo}}$. Similarly, $\chi^\flat{}_\perp$ restricts $\vec{\hat{\pi}}^\flat{}_{\overline{k}}$ to a solenoidal axial vector, removing 1 D.o.F. A further D.o.F is removed by substituting $\mathcal{H}^\flat{}_{\overline{kl}}$ into $\mathcal{H}^\flat{}_\alpha$, and a final D.o.F is removed by $\widetilde{\chi}^\flat{}_{\perp\overline{kl}}$. Separately, $\mathcal{H}^\flat{}_{\overline{kl}}$ removes 3 D.o.Fs. All the PiCs and SiCs are FC, and 1 D.o.F remains, as expected from Table I

$$\begin{aligned}1 = \frac{1}{2}\big(&80 - 2\times 10[\text{sPFC}] - 2\times(1+3+2)[\text{sSFC}]\\ &- 2\times(1+3+5+1+5+5)[\text{iPFC}]\\ &- 2\times(1+1+1)[\text{iSFC}]\big).\end{aligned} \tag{28}$$

C. Case *5 25

The PPM is:

$$\begin{array}{c|c|c|c|c|c|c|c|}
 & \widetilde{\varphi}_{\overline{kl}} & \varphi_\perp & \hat{\varphi}_{\perp\overline{kl}} & \vec{\varphi}_{\overline{k}} & \widetilde{\varphi}_{\perp\overline{kl}} & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \\
\hline
\widetilde{\varphi}_{\overline{kl}} & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & \hat{\pi} & 5\\
\hline
\varphi_\perp & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1\\
\hat{\varphi}_{\perp\overline{kl}} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 3\\
\vec{\varphi}_{\overline{k}} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 3\\
\widetilde{\varphi}_{\perp\overline{kl}} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 5\\
{}^{\mathrm{T}}\varphi_{\overline{klo}} & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & \cdot & 5\\
\hline
 & 5 & 1 & 3 & 3 & 5 & 5 &
\end{array} \tag{29}$$

Within the PiC shell, we have

$$\widetilde{\chi}^\flat{}_{\overline{kl}} \approx -\mathcal{D}^\flat{}_{\langle\overline{k}}\hat{\pi}^\flat{}_{\perp\overline{l}\rangle} \tag{30a}$$

$$\chi^\flat{}_\perp \approx \frac{1}{2}\hat{\pi}^\flat, \tag{30b}$$

$$\hat{\chi}^\flat{}_{\perp\overline{kl}} \approx \hat{\hat{\pi}}^\flat{}_{\overline{kl}} - \frac{1}{12}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^\flat, \tag{30c}$$

$$\vec{\chi}^\flat{}_{\overline{k}} \approx \hat{\pi}^\flat{}_{\perp\overline{k}}, \tag{30d}$$

and this time, all 10 sSFCs persist in the PiC shell

$$\mathcal{H}^\flat{}_\perp \approx -\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\pi}^\flat{}_{\perp\overline{l}} \tag{31a}$$

$$\mathcal{H}^\flat{}_\alpha \approx -\frac{1}{3}h^\flat{}_\alpha{}^{\overline{k}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\pi}^\flat - h^\flat{}_\alpha{}^{\overline{k}}\eta^{\flat\overline{jl}}\mathcal{D}^\flat{}_{\overline{j}}\hat{\hat{\pi}}^\flat{}_{\overline{kl}}, \tag{31b}$$

$$\mathcal{H}^\flat{}_{\overline{kl}} \approx 2\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - \frac{1}{6}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^\flat, \tag{31c}$$

$$\mathcal{H}^\flat{}_{\perp\overline{k}} \approx \hat{\pi}^\flat{}_{\perp\overline{k}}. \tag{31d}$$

We find that $\mathcal{H}^\flat{}_{\perp\overline{k}}$ and $\mathcal{H}^\flat{}_{\overline{kl}}$ each remove 3 D.o.F, while $\chi^\flat{}_\perp$ removes 1 D.o.F; the remaining sSFCs and SiCs are then implied, and the PiCs and SiCs are FC. Once again, 1 D.o.F remains as expected from Table I

$$\begin{aligned}1 = \frac{1}{2}\big(&80 - 2\times 10[\text{sPFC}] - 2\times(3+3)[\text{sSFC}]\\ &- 2\times(5+1+3+3+5+5)[\text{iPFC}]\\ &- 2\times 1[\text{iSFC}]\big).\end{aligned} \tag{32}$$

D. Case 24

The PPM is:

$$\begin{array}{c|c|c|c|c|c|c|c|}
 & \widetilde{\varphi}_{\overline{kl}} & \varphi_\perp & \widetilde{\varphi}_{\perp\overline{kl}} & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \\
\hline
\widetilde{\varphi}_{\overline{kl}} & \hat{\pi} & \cdot & \hat{\pi} & \hat{\pi} & 5\\
\hline
\varphi_\perp & \cdot & \cdot & \cdot & \cdot & 1\\
\widetilde{\varphi}_{\perp\overline{kl}} & \hat{\pi} & \cdot & \cdot & \cdot & 5\\
{}^{\mathrm{T}}\varphi_{\overline{klo}} & \hat{\pi} & \cdot & \cdot & \cdot & 5\\
\hline
 & 5 & 1 & 5 & 5 &
\end{array} \tag{33}$$

Within the PiC shell, we have the following SiCs

$$\widetilde{\chi}^\flat{}_{\overline{kl}} \approx -\mathcal{D}^\flat{}_{\langle\overline{k}}\hat{\pi}^\flat{}_{\perp\overline{l}\rangle} \tag{34a}$$

$$\chi^\flat{}_\perp \approx \frac{1}{2}\hat{\pi}^\flat - \frac{1}{2}\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\overline{k}}\vec{\hat{\pi}}^\flat{}_{\overline{l}}, \tag{34b}$$

$$\widetilde{\chi}^\flat{}_{\perp\overline{kl}} \approx \frac{1}{4}\mathcal{D}^\flat{}_{\langle\overline{k}}\vec{\hat{\pi}}^\flat{}_{\overline{l}\rangle}, \tag{34c}$$

$$\begin{aligned}{}^{\mathrm{T}}\chi^\flat{}_{\overline{klo}} \approx\ &\frac{1}{4}\mathcal{D}^\flat{}_{\overline{o}}\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}} + \frac{1}{4}\mathcal{D}^\flat{}_{[\overline{l}}\hat{\hat{\pi}}^\flat{}_{\perp\overline{k}]\overline{o}}\\ &+ \frac{3}{8}\eta^\flat{}_{\overline{o}[\overline{k}|}\eta^{\flat\overline{ij}}\mathcal{D}^\flat{}_{\overline{i}}\hat{\hat{\pi}}^\flat{}_{\perp|\overline{l}]\overline{j}},\end{aligned} \tag{34d}$$

and the following sPFCs

$$\mathcal{H}^\flat{}_\perp \approx -\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\pi}^\flat{}_{\perp\overline{l}} \tag{35a}$$

$$\mathcal{H}^\flat{}_\alpha \approx -\frac{1}{3}h^\flat{}_\alpha{}^{\overline{k}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\pi}^\flat - h^\flat{}_\alpha{}^{\overline{k}}\eta^{\flat\overline{jl}}\mathcal{D}^\flat{}_{\overline{j}}\hat{\hat{\pi}}^\flat{}_{\overline{kl}}, \tag{35b}$$

$$\mathcal{H}^\flat{}_{\overline{kl}} \approx 2\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - \frac{1}{6}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^\flat + \mathcal{D}^\flat{}_{[\overline{k}}\vec{\hat{\pi}}^\flat{}_{\overline{l}]}, \tag{35c}$$

$$\mathcal{H}^\flat{}_{\perp\overline{k}} \approx \hat{\pi}^\flat{}_{\perp\overline{k}} + \eta^{\flat\overline{jl}}\mathcal{D}^\flat{}_{\overline{j}}\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}}. \tag{35d}$$

It is clear from the PiC shell that Case 24 has much in common with Case 28, and again we will implement the SiCs and sSFCs simultaneously. Firstly, we find that ${}^{\mathrm{T}}\chi^\flat{}_{\overline{klo}}$ constitutes an overdetermined system in $\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}}$, which vanishes and takes with it 3 D.o.F. Consequently, from $\mathcal{H}^\flat{}_{\perp\overline{k}}$, we see that $\hat{\pi}^\flat{}_{\perp\overline{k}}$ must vanish along with another 3 D.o.F, such that $\mathcal{H}^\flat{}_\perp$ and $\widetilde{\chi}^\flat{}_{\overline{kl}}$ vanish automatically. Similarly, $\widetilde{\chi}^\flat{}_{\perp\overline{kl}}$ is an overdetermined system in $\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}}$, which vanishes with another 3 D.o.F; $\chi^\flat{}_\perp$ then causes $\hat{\pi}^\flat$ to vanish with 1 D.o.F. As before, 1 D.o.F propagates in accordance with Table I

$$\begin{aligned}1 = \frac{1}{2}\big(&80 - 2\times 10[\text{sPFC}] - 2\times(3+3)[\text{sSFC}]\\ &- 2\times(5+1+5+5)[\text{iPFC}]\\ &- 2\times(1+3+3)[\text{iSFC}]\big).\end{aligned} \tag{36}$$

E. Case 32

For the first time, we encounter non-vanishing mass parameters in the PiCs, specifically in $\hat{\hat{\varphi}}^\flat{}_{\perp\overline{kl}}$ and ${}^{\mathrm{T}}\varphi^\flat{}_{\overline{klo}}$. We anticipate the nonvanishing commutators

$\{\hat{\varphi}^\flat{}_{\perp\overline{kl}}, \hat{\chi}^\flat{}_{\perp\overline{ij}}\} \approx \mathcal{O}(1)$ – noting that the natural conjugate $\hat{\varphi}^\flat{}_{\overline{kl}}$ is not a PiC – and $\{{}^{\mathrm{T}}\varphi^\flat{}_{\overline{klo}}, {}^{\mathrm{T}}\chi^\flat{}_{\overline{ijn}}\} \approx \mathcal{O}(1)$. These PiCs and SiCs will be SC, allowing for the determination of their multipliers. The PPM of this theory is remarkable, since it remains empty even in the nonlinear regime:

$$
\begin{array}{cc|cccccc|c}
 & & & & \downarrow & & & \downarrow & \\
 & & \varphi & \varphi_\perp & \hat{\varphi}_{\perp\overline{kl}} & \vec{\varphi}_{\overline{k}} & \tilde{\varphi}_{\perp\overline{kl}} & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \\
\hline
 & \varphi & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 \\
 & \varphi_\perp & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 \\
\rightarrow & \hat{\varphi}_{\perp\overline{kl}} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 3 \\
 & \vec{\varphi}_{\overline{k}} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 3 \\
 & \tilde{\varphi}_{\perp\overline{kl}} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 5 \\
\rightarrow & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 5 \\
\hline
 & & 1 & 1 & 3 & 3 & 5 & 5 &
\end{array}
\tag{37}
$$

Within the PiC shell, we find the following SiCs

$$
\chi^\flat \approx \eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\pi}^\flat{}_{\perp\overline{l}}, \tag{38a}
$$

$$
\begin{aligned}
\hat{\chi}^\flat{}_{\perp\overline{kl}} \approx & -\frac{\hat{\beta}_1 + 2\hat{\beta}_3}{2(\hat{\beta}_1 - \hat{\beta}_3)}\hat{\pi}^\flat{}_{\overline{kl}} \\
& - \frac{1}{12}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^\flat \\
& + \frac{9\hat{\beta}_1\hat{\beta}_3}{2(\hat{\beta}_1 - \hat{\beta}_3)(\hat{\beta}_1 + 2\hat{\beta}_3)}\hat{\varphi}^\flat{}_{\overline{kl}},
\end{aligned} \tag{38b}
$$

$$
\vec{\chi}^\flat{}_{\overline{k}} \approx \hat{\pi}^\flat{}_{\perp\overline{k}}, \tag{38c}
$$

$$
\tilde{\chi}^\flat{}_{\perp\overline{kl}} \approx \tilde{\hat{\pi}}^\flat{}_{\overline{kl}}, \tag{38d}
$$

$$
{}^{\mathrm{T}}\chi^\flat{}_{\overline{klo}} \approx 2\hat{\beta}_1\,{}^{\mathrm{T}}\mathcal{T}^\flat{}_{\overline{klo}}\,. \tag{38e}
$$

Note the appearance of field strengths, specifically the torsion in $\hat{\varphi}^\flat{}_{\overline{kl}}$ and ${}^{\mathrm{T}}\mathcal{T}^\flat{}_{\overline{klo}}$. Whilst these somewhat complicate the analysis, they naturally appear with the masses. We also mark the first apparent instance of a TiC accompanying $\tilde{\chi}^\flat{}_{\perp\overline{kl}}$. Using the notation $\zeta = \dot{\chi}$, this may be written as

$$
\tilde{\zeta}^\flat{}_{\perp\overline{kl}} \approx \frac{4}{3}\eta^{\flat\overline{ij}}\mathcal{D}^\flat{}_{\overline{i}}\,{}^{\mathrm{T}}\chi^\flat{}_{\langle\overline{k}|\overline{j}|\overline{l}\rangle}, \tag{39}
$$

which then vanishes in the SiC shell. The PiC shell contains the following sSFCs:

$$
\mathcal{H}^\flat{}_\perp \approx -\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\overline{k}}\hat{\pi}^\flat{}_{\perp\overline{l}} \tag{40a}
$$

$$
\mathcal{H}^\flat{}_\alpha \approx -h^\flat{}_\alpha{}^{\overline{k}}\eta^{\flat\overline{jl}}\mathcal{D}^\flat{}_{\overline{j}}\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - h^\flat{}_\alpha{}^{\overline{k}}\eta^{\flat\overline{jl}}\mathcal{D}^\flat{}_{\overline{j}}\tilde{\hat{\pi}}^\flat{}_{\overline{kl}}, \tag{40b}
$$

$$
\mathcal{H}^\flat{}_{\overline{kl}} \approx 2\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - \frac{1}{6}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^\flat, \tag{40c}
$$

$$
\mathcal{H}^\flat{}_{\perp\overline{k}} \approx \hat{\pi}^\flat{}_{\perp\overline{k}}. \tag{40d}
$$

Since two of the PiC chains are known to be self-terminating, the algorithm concludes quite quickly. As with Case [*5]25, $\mathcal{H}^\flat{}_{\perp\overline{k}}$ and $\mathcal{H}^\flat{}_{\overline{kl}}$ each eliminate 3 D.o.F. Another 5 D.o.F are then removed by $\tilde{\chi}^\flat{}_{\perp\overline{kl}}$, with the remaining SiCs and sSFCs automatically satisfied. The 1 remaining D.o.F is again expected from Table I

$$
\begin{aligned}
1 = \frac{1}{2}\big(&80 - 2\times 10[\text{sPFC}] - 2\times(3+3)[\text{sSFC}] \\
&- 2\times(1+1+3+5)[\text{iPFC}] - (3+5)[\text{iPSC}] \\
&- 2\times 5[\text{iSFC}] - (3+5)[\text{iSSC}]\big).
\end{aligned} \tag{41}
$$

F. Case 20

The analysis of Case 20 is quite similar to Case 32. Mass parameters again accompany the PiCs, and we expect $\hat{\varphi}^\flat{}_{\perp\overline{kl}}$, $\vec{\varphi}^\flat{}_{\overline{k}}$ and ${}^{\mathrm{T}}\varphi^\flat{}_{\overline{kl\rho}}$ to not commute with their respective SiCs on the final shell. Once again, the PPM is empty both before and after linearisation:

$$
\begin{array}{cc|ccccc|c}
 & & & \downarrow & \downarrow & & \downarrow & \\
 & & \varphi_\perp & \hat{\varphi}_{\perp\overline{kl}} & \vec{\varphi}_{\overline{k}} & \tilde{\varphi}_{\perp\overline{kl}} & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \\
\hline
 & \varphi_\perp & \cdot & \cdot & \cdot & \cdot & \cdot & 1 \\
\rightarrow & \hat{\varphi}_{\perp\overline{kl}} & \cdot & \cdot & \cdot & \cdot & \cdot & 3 \\
\rightarrow & \vec{\varphi}_{\overline{k}} & \cdot & \cdot & \cdot & \cdot & \cdot & 3 \\
 & \tilde{\varphi}_{\perp\overline{kl}} & \cdot & \cdot & \cdot & \cdot & \cdot & 5 \\
\rightarrow & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \cdot & \cdot & \cdot & \cdot & \cdot & 5 \\
\hline
 & & 1 & 3 & 3 & 5 & 5 &
\end{array}
\tag{42}
$$

Within the PiC shell, we first the following SiCs

$$
\chi^\flat{}_\perp \approx \frac{1}{2}\hat{\pi}^\flat, \tag{43a}
$$

$$
\begin{aligned}
\hat{\chi}^\flat{}_{\perp\overline{kl}} \approx & -\frac{\hat{\beta}_1 + 2\hat{\beta}_3}{2(\hat{\beta}_1 - \hat{\beta}_3)}\hat{\hat{\pi}}^\flat{}_{\overline{kl}} \\
& - \frac{1}{12}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\overline{n}}{}^{\mathrm{P}}\hat{\pi}^\flat \\
& + \frac{9\hat{\beta}_1\hat{\beta}_3}{2(\hat{\beta}_1 - \hat{\beta}_3)(\hat{\beta}_1 + 2\hat{\beta}_3)}\hat{\varphi}^\flat{}_{\overline{kl}},
\end{aligned} \tag{43b}
$$

$$
\vec{\chi}^\flat{}_{\overline{k}} \approx -\frac{\hat{\beta}_1 + 2\hat{\beta}_2}{2(\hat{\beta}_1 - \hat{\beta}_2)}\hat{\pi}^\flat{}_{\perp\overline{k}} \tag{43c}
$$

$$
+ \frac{9\hat{\beta}_1\hat{\beta}_2}{2(\hat{\beta}_1 - \hat{\beta}_2)(2\hat{\beta}_1 + \hat{\beta}_2)}\varphi^\flat{}_{\perp\overline{k}}, \tag{43d}
$$

$$
\tilde{\chi}^\flat{}_{\perp\overline{kl}} \approx \frac{1}{2}\tilde{\hat{\pi}}^\flat{}_{\overline{kl}}, \tag{43e}
$$

$$
{}^{\mathrm{T}}\chi^\flat{}_{\overline{klo}} \approx 2\hat{\beta}_1\,{}^{\mathrm{T}}\mathcal{T}^\flat{}_{\overline{klo}}\,. \tag{43f}
$$

This time, two TiCs appear, but uppon rearranging both may eventually be written in terms of the iSSCs, and are satisfied automatically

$$
\zeta^\flat{}_\perp \approx \eta^{\flat\overline{ij}}\mathcal{D}^\flat{}_{\overline{i}}\vec{\chi}^\flat{}_{\overline{j}}, \tag{44a}
$$

$$
\tilde{\zeta}^\flat{}_{\perp\overline{kl}} \approx \frac{4}{3}\eta^{\flat\overline{ij}}\mathcal{D}^\flat{}_{\overline{i}}\,{}^{\mathrm{T}}\chi^\flat{}_{\langle\overline{k}|\overline{j}|\overline{l}\rangle} - \frac{1}{2}\mathcal{D}^\flat{}_{\langle\overline{k}}\vec{\chi}^\flat{}_{\overline{l}\rangle}. \tag{44b}
$$

The sSFC content in the PiC shell is largely the same as that of Case 32, with the only difference marked in the

linear super-momentum

$$\mathcal{H}^\flat{}_\alpha \approx -\frac{1}{3}h^\flat{}_\alpha{}^{\bar{k}}\mathcal{D}^\flat{}_{\bar{k}}\hat{\pi}^\flat - h^\flat{}_\alpha{}^{\bar{k}}\eta^{\flat\bar{j}\bar{l}}\mathcal{D}^\flat{}_{\bar{j}}\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - h^\flat{}_\alpha{}^{\bar{k}}\eta^{\flat\bar{j}\bar{l}}\mathcal{D}^\flat{}_{\bar{j}}\tilde{\hat{\pi}}^\flat{}_{\overline{kl}}. \tag{45a}$$

Aided by the additional conjugate pair of constraints, the algorithm terminates even faster than with Case 32: we see that 1 and 5 D.o.F are removed by each of $\chi^\flat{}_\perp$ and $\tilde{\varphi}^\flat{}_{\perp\overline{kl}}$. As before, the 1 propagating D.o.F is confirmed from Table I

$$\begin{aligned}1 = \frac{1}{2}\big(&80 - 2\times 10[\text{sPFC}] - 2\times(3+3)[\text{sSFC}]\\ &- 2\times(1+5)[\text{iPFC}] - (3+3+5)[\text{iPSC}]\\ &- 2\times(1+5)[\text{iSFC}] - (3+3+5)[\text{iSSC}]\big).\end{aligned} \tag{46}$$

IV. MASSLESS THEORIES

A. Case 17

As mentioned in Section II B, the evaluation of the PPM is complicated by the appearance of torsion in PiCs belonging to the translational sector, which generates derivatives of the Dirac function. Fortunately, these derivatives may be discarded up to a surface term even within the PiC shell, and the entire matrix may be written in terms of parallel momenta as before

$$\begin{array}{cc|cccccc|c}
 & & & \downarrow & & \downarrow & & \downarrow & \\
 & & \varphi & \hat{\varphi}_{\overline{kl}} & \varphi_\perp & {}^{\text{P}}\varphi & \tilde{\varphi}_{\perp\overline{kl}} & {}^{\text{T}}\varphi_{\overline{klo}} & \\
\hline
 & \varphi & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 1\\
\to & \hat{\varphi}_{\overline{kl}} & \hat{\pi} & \cdot & \hat{\pi} & \hat{\pi} & \hat{\pi} & \hat{\pi} & 3\\
 & \varphi_\perp & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 1\\
\to & {}^{\text{P}}\varphi & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 1\\
 & \tilde{\varphi}_{\perp\overline{kl}} & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 5\\
\to & {}^{\text{T}}\varphi_{\overline{klo}} & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 5\\
\hline
 & & 1 & 3 & 1 & 1 & 5 & 5 &
\end{array} \tag{47}$$

Due to the appearance of mass parameters, we will expect $\hat{\varphi}^\flat{}_{\overline{kl}}$, ${}^{\text{P}}\varphi^\flat$ and ${}^{\text{T}}\varphi^\flat{}_{\overline{klo}}$ not to commute with their SiCs in the final shell. Within the PiC shell, we find the following SiCs

$$\chi^\flat \approx -\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\bar{k}}\hat{\pi}^\flat{}_{\perp\bar{l}}, \tag{48a}$$

$$\hat{\chi}^\flat{}_{\overline{kl}} \approx -\frac{\hat{\beta}_3}{\hat{\alpha}_5}\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}} - \mathcal{D}^\flat{}_{[\bar{k}}\hat{\pi}^\flat{}_{\perp\bar{l}]} + 2\hat{\beta}_3\eta^{\flat\bar{i}\bar{j}}\mathcal{D}^\flat{}_{\bar{i}}\mathcal{T}^\flat{}_{\overline{jkl}} - 4\hat{\beta}_3\mathcal{D}^\flat{}_{[\bar{k}}\vec{\mathcal{T}}^\flat{}_{\bar{l}]}, \tag{48b}$$

$$\chi^\flat{}_\perp \approx -\frac{1}{2}\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\bar{k}}\vec{\hat{\pi}}^\flat{}_{\bar{l}}, \tag{48c}$$

$${}^{\text{P}}\chi^\flat \approx \epsilon^{\flat\overline{jkl}\perp}\mathcal{D}^\flat{}_{\bar{j}}\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}} - 4\hat{\beta}_3{}^{\text{P}}\mathcal{T}^\flat, \tag{48d}$$

$$\tilde{\chi}^\flat{}_{\perp\overline{kl}} \approx \frac{1}{2}\tilde{\hat{\pi}}^\flat{}_{\overline{kl}} + \frac{1}{4}\mathcal{D}^\flat{}_{\langle\bar{k}}\vec{\hat{\pi}}^\flat{}_{\bar{l}\rangle}, \tag{48e}$$

$${}^{\text{T}}\chi^\flat{}_{\overline{klo}} \approx \frac{1}{4}\mathcal{D}^\flat{}_{\bar{o}}\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}} + \frac{1}{4}\mathcal{D}^\flat{}_{[\bar{l}}\hat{\hat{\pi}}^\flat{}_{\perp\bar{k}]\bar{o}} + \frac{3}{8}\eta^\flat{}_{\bar{o}[\bar{k}|}\eta^{\flat\bar{i}\bar{j}}\mathcal{D}^\flat{}_{\bar{i}}\hat{\hat{\pi}}^\flat{}_{\perp|\bar{l}]\bar{j}} - 4\hat{\beta}_3{}^{\text{T}}\mathcal{T}^\flat{}_{\overline{klo}}. \tag{48f}$$

Among these, we note that a TiC accompanies $\tilde{\chi}^\flat{}_{\perp\overline{kl}}$, but may be expressed in terms of ${}^{\text{T}}\chi^\flat{}_{\overline{klo}}$ by precisely (39). Within the PiC shell, the sSFCs are

$$\mathcal{H}^\flat{}_\perp \approx -\eta^{\flat\overline{kl}}\mathcal{D}^\flat{}_{\bar{k}}\hat{\pi}^\flat{}_{\perp\bar{l}}, \tag{49a}$$

$$\mathcal{H}^\flat{}_\alpha \approx -h^\flat{}_\alpha{}^{\bar{k}}\eta^{\flat\bar{j}\bar{l}}\mathcal{D}^\flat{}_{\bar{j}}\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - h^\flat{}_\alpha{}^{\bar{k}}\eta^{\flat\bar{j}\bar{l}}\mathcal{D}^\flat{}_{\bar{j}}\tilde{\hat{\pi}}^\flat{}_{\overline{kl}}, \tag{49b}$$

$$\mathcal{H}^\flat{}_{\overline{kl}} \approx 2\hat{\hat{\pi}}^\flat{}_{\overline{kl}} + \mathcal{D}^\flat{}_{[\bar{k}}\vec{\hat{\pi}}^\flat{}_{\bar{l}]}, \tag{49c}$$

$$\mathcal{H}^\flat{}_{\perp\bar{k}} \approx \hat{\pi}^\flat{}_{\perp\bar{k}} + \eta^{\flat\bar{j}\bar{l}}\mathcal{D}^\flat{}_{\bar{j}}\hat{\hat{\pi}}^\flat{}_{\perp\overline{kl}}. \tag{49d}$$

The conjugate pairs together eliminate 6, 2 and 10 D.o.F before terminating. As with Case 28, $\hat{\pi}^\flat{}_{\bar{k}}$ becomes solenoidal due to $\chi^\flat{}_\perp$ and loses 1 D.o.F, while 3 D.o.F are lost by each of $\mathcal{H}^\flat{}_{\overline{kl}}$ and $\mathcal{H}^\flat{}_{\perp\bar{k}}$. In total, 2 propagating D.o.F remain as expected from Table I

$$\begin{aligned}2 = \frac{1}{2}\big(&80 - 2\times 10[\text{sPFC}] - 2\times(3+3)[\text{sSFC}]\\ &- 2\times(1+1+5)[\text{iPFC}] - (3+1+5)[\text{iPSC}]\\ &- 2\times(1+5)[\text{iSFC}] - (3+1+5)[\text{iSSC}]\big).\end{aligned} \tag{50}$$

B. Case 3

It should come as no surprise that Case 3 is a relaxation of Case 17, which admits an extra D.o.F. The PPM is:

$$\begin{array}{cc|ccccc|c}
 & & & \downarrow & & & \downarrow & \\
 & & \varphi & \hat{\varphi}_{\overline{kl}} & \varphi_\perp & \tilde{\varphi}_{\perp\overline{kl}} & {}^{\text{T}}\varphi_{\overline{klo}} & \\
\hline
 & \varphi & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & 1\\
\to & \hat{\varphi}_{\overline{kl}} & \hat{\pi} & \cdot & \hat{\pi} & \hat{\pi} & \hat{\pi} & 3\\
 & \varphi_\perp & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & 1\\
 & \tilde{\varphi}_{\perp\overline{kl}} & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & 5\\
\to & {}^{\text{T}}\varphi_{\overline{klo}} & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & 5\\
\hline
 & & 1 & 3 & 1 & 5 & 5 &
\end{array} \tag{51}$$

It is clear that ${}^{\text{P}}\varphi^\flat$ is no longer primarily constrained. The only change to the remaining SiCs of Case 17 is

$$\mathcal{H}^\flat{}_{\overline{kl}} \approx 2\hat{\hat{\pi}}^\flat{}_{\overline{kl}} - \frac{1}{6}\epsilon^\flat{}_{\overline{klm}\perp}\eta^{\flat\overline{mn}}\mathcal{D}^\flat{}_{\bar{n}}{}^{\text{P}}\hat{\pi}^\flat + \mathcal{D}^\flat{}_{[\bar{k}}\vec{\hat{\pi}}^\flat{}_{\bar{l}]}, \tag{52}$$

but $\mathcal{H}^\flat{}_\alpha$ is still satisfied. Overall, only the conjugate ${}^{\text{P}}\varphi^\flat$ and ${}^{\text{P}}\chi^\flat$ pair are removed, leaving 3 propagating D.o.F as expected from Table I

$$\begin{aligned}3 = \frac{1}{2}\big(&80 - 2\times 10[\text{sPFC}] - 2\times(3+3)[\text{sSFC}]\\ &- 2\times(1+1+5)[\text{iPFC}] - (3+5)[\text{iPSC}]\\ &- 2\times(1+5)[\text{iSFC}] - (3+5)[\text{iSSC}]\big).\end{aligned} \tag{53}$$

V. PHENOMENOLOGY

$$
\begin{array}{c|cccccc|c}
 & & \downarrow & & \downarrow & & \downarrow & \\
 & \varphi & \hat{\varphi}_{\overline{kl}} & \varphi_{\perp} & {}^{\mathrm{P}}\varphi & \widetilde{\varphi}_{\perp\overline{kl}} & {}^{\mathrm{T}}\varphi_{\overline{klo}} & \\
\hline
\varphi & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 1 \\
\rightarrow \hat{\varphi}_{\overline{kl}} & \hat{\pi} & \cdot & \hat{\pi} & \hat{\pi} & \hat{\pi} & \hat{\pi} & 3 \\
\varphi_{\perp} & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 1 \\
\rightarrow {}^{\mathrm{P}}\varphi & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 1 \\
\widetilde{\varphi}_{\perp\overline{kl}} & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 5 \\
\rightarrow {}^{\mathrm{T}}\varphi_{\overline{klo}} & \cdot & \hat{\pi} & \cdot & \cdot & \cdot & \cdot & 5 \\
\hline
 & 1 & 3 & 1 & 1 & 5 & 5 &
\end{array}
\tag{54}
$$

Will put a quick scan of the metrical analogues of Case 3 and Case 17 here, and possibly citation of Anthony's results (if appropriate wrt time of publishing etc.).

VI. CONCLUSIONS

Some theories are possibly healthy. Or possibly, they are not healthy. Clearly we have learned much.

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Appendix A: Lorentz projections

Appendix B: Nonlinear Poisson brackets

1. Case ^{*6}26

2. Case 28

3. Case ^{*5}25

4. Case 24

5. Case 17

6. Case 32

7. Case 3

[1] A. N. Lasenby and M. P. Hobson, Journal of Mathematical Physics **57**, 092505 (2016), arXiv:1510.06699 [gr-qc].

[2] W. E. V. Barker, A. N. Lasenby, M. P. Hobson, and W. J. Handley, Phys. Rev. D **102**, 024048 (2020), arXiv:2003.02690 [gr-qc].